

① $V - V - F - V$

②

(a) $0 \times 5 \text{ Cd (hex-bin)}$ 0000 0101 1100 0010₂
 | | |
 | | 0010
 | 1100
 0101

(b) $2048 \cdot 10^6$

$$\frac{2048 \ 000 \ 000}{1 \ 073 \ 241 \ 824} = 1.907 \text{ GIB}$$

(c) 10110100
 - 01001101

 11100111

③ (a) -57 (dec-bin)

$$\begin{array}{r} 57 \\ - 32 \\ \hline 25 \\ - 16 \\ \hline 9 \\ - 8 \\ \hline 1 \end{array}$$

$$\cancel{128} \ \cancel{64} \ \textcircled{32} \ \textcircled{16} \ \textcircled{8} \ \cancel{4} \ \cancel{2} \ \cancel{1}$$

$$00111001$$

$$0111000110_2$$

(b) $10011010_2 \text{ (bin-dec)}$

$$01100101$$

$$\begin{array}{l} | \\ | \\ | \\ | \\ | \\ | \\ | \end{array}$$

$$\begin{array}{l} 1 \cdot 2^0 = 1 \\ 0 \cdot 2^1 = 0 \\ 1 \cdot 2^2 = 4 \\ 0 \cdot 2^3 = 0 \\ 0 \cdot 2^4 = 0 \\ 1 \cdot 2^5 = 32 \\ 1 \cdot 2^6 = 64 \\ 0 \cdot 2^7 = 0 \end{array}$$

$$\begin{array}{r} 64 \\ + 32 \\ \hline 96 \\ + 5 \\ \hline 101 \end{array}$$

$$= 101_{10}$$

③ $01010000 + 01110000 =$

$$\begin{array}{r} 01010000 \\ + 01110000 \\ \hline 11000000 \end{array} \rightarrow \text{overflow}$$

④

② CP_{global}

$$CP_g = \frac{10^6 \cdot (400 \cdot 1 + 100 \cdot 3 + 150 \cdot 4 + 50 \cdot 4)}{10^6 \cdot (400 + 100 + 150 + 50)} = \frac{\cancel{10^6} \cdot 1500}{\cancel{10^6} \cdot 700} = 2,142$$

$$b) T_{cpu} = \frac{1500 \cdot 10^6}{2,5 \cdot 10^8} = \frac{\cancel{1500} \cdot \cancel{10^6}}{\cancel{2500} \cdot \cancel{10^6}} = 0,6 \mu$$

$$c) CP_1 = \frac{10^6 \cdot (400 \cdot 1 + 100 \cdot 3 + 75 \cdot 4 + 50 \cdot 4)}{10^6 \cdot (400 + 100 + 75 + 50)} = \frac{1200}{625} = 1,92$$

$$T_{cpu} = \frac{1200 \cdot 10^6}{2,5 \cdot 10^8} = \frac{\cancel{1200} \cdot \cancel{10^6}}{\cancel{2500} \cdot \cancel{10^6}} = 0,48 \mu \quad \text{speedup} = \frac{0,6}{0,48} = 1,25x$$

⑤

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int i = 0;
do {
    S1 = n
    C[i] = A[i] + k;
    S2 = 2A
    S3 = k
    S4 = 2C
    i++;
} while (i < n);

```

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limits:

addi \$sp, \$sp, -8

sw \$ra, 0(\$sp)

add \$t0, \$a0, \$a2

sw \$a0, 4(\$sp)

add \$a0, \$zero, \$t0

jal menu

sub \$v0, \$v0, \$a2

sw \$ra, 0(\$sp)

sw \$a0, 4(\$sp)

jr \$ra

7 instruções de sw, verificar se camadas do professor